MATH 2413 - Prior Test (SP24)
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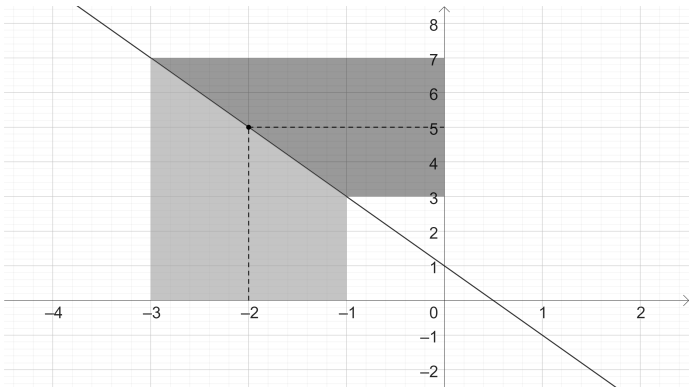
Test 1

Directions:
- Turn off your cell phone **now**.
- Calculators ARE allowed during the test.
- No notes or external scratch paper allowed. Scratch paper is provided in the back; you may rip it out if you wish, but you must submit it with the test.
- Read all of the directions for each question.
- Answer the questions to the best of your ability in the space provided.
- Show **all** of your work unless otherwise noted.
- **Box** your answers.

Question:	1	2	3	4	5	6	Total
Points:	18	16	16	16	18	16	100
Score:							

1. (a) (6 points) The graph below is a partially labeled diagram depicting an $\varepsilon - \delta$ limit of a linear function. Write out the limit in the form $\lim_{x \to c} f(x) = L$ with the correct c, $f(x)$, and L in place. Also, determine the values of ε and δ shown in the graph. *No need to show work for this part.*



Solution: The equation of the line can be found to be $y = -2x + 1$. So, our limit with ε and δ is

$$\lim_{x \to -2} -2x + 1 = 5, \quad \varepsilon = 2, \quad \delta = 1$$

(b) (6 points) For this limit, find δ in terms of ε generally. *Note: we are no longer looking at the specific values of ε and δ from part a). This is the "scratch work" part of a $\varepsilon - \delta$ proof.*

Solution: Using the definition in reverse, we have

$$|(-2x+1) - (5)| < \varepsilon \implies |-2x - 4| < \varepsilon$$
$$\implies |-2||x+2| < \varepsilon \implies |x+2| < \frac{\varepsilon}{2} = \delta$$

(c) (6 points) Write a formal proof for the limit. Start by filling in the blanks, then proceeding using algebra.

Let $\varepsilon >$ _____ and $\delta =$ _____. Then, we have...

Solution: Let $\varepsilon > 0$ and $\delta = \frac{\varepsilon}{2}$. We have,

$$0 < |x+2| < \delta \implies |x+2| < \frac{\varepsilon}{2}$$
$$\implies 2|x+2| < \varepsilon \implies |-2||x+2| < \varepsilon$$
$$\implies |-2x-4| < \varepsilon \implies |(-2x+1) - (5)| < \varepsilon$$

So, $\lim_{x \to -2} -2x + 1 = 5$.

2. Compute the following limits. Note: You may not use any limit rules that involve derivatives for this problem (i.e. L'Hopital's Rule).

   (a) (4 points) $\lim_{x\to 2} \frac{\log_2(x)}{x-1}$

   Solution: We have

$$\lim_{x\to 2} \frac{\log_2(x)}{x-1} = \frac{1}{1} = 1$$

   (b) (6 points) $\lim_{x\to 3} \frac{x^3-9x}{x^2-3x}$

   Solution: Factoring, we have

$$\lim_{x\to 3} \frac{x^3-9x}{x^2-3x} = \lim_{x\to 3} \frac{x(x-3)(x+3)}{x(x-3)}$$
$$= \lim_{x\to 3} \frac{1}{x+3} = \frac{1}{6}$$

   (c) (6 points) $\lim_{x\to 1^+} \frac{\frac{1}{x}-1}{x-1}$

   Solution: We have

$$\lim_{x\to 1^+} \frac{\frac{1}{x}-1}{x-1}\left(\frac{x}{x}\right)$$
$$\lim_{x\to 1^+} \frac{1-x}{x(x-1)}$$
$$\lim_{x\to 1^+} \frac{-(x-1)}{x(x-1)} = \lim_{x\to 1^+} \frac{-1}{x} = -1$$

3. Consider the function $f(x) = \dfrac{-x}{x^2 - 1}$.

(a) (8 points) Find all horizontal asymptotes of $f(x)$. Support your answers with limit statements.

Solution: We consider the limits of the function as $x \to \pm\infty$.

$$\lim_{x\to\infty} \frac{-x}{x^2-1}\left(\frac{1/x^2}{1/x^2}\right) = \lim_{x\to\infty} \frac{-1/x}{1-1/x^2} = 0$$

Similarly,

$$\lim_{x\to-\infty} \frac{-x}{x^2-1}\left(\frac{1/x^2}{1/x^2}\right) = \lim_{x\to-\infty} \frac{-1/x}{1-1/x^2} = 0$$

So, there is a horizontal asymptote at $y = 0$.

(b) (8 points) Find all vertical asymptotes of $f(x)$. Support your answers with limit statements.

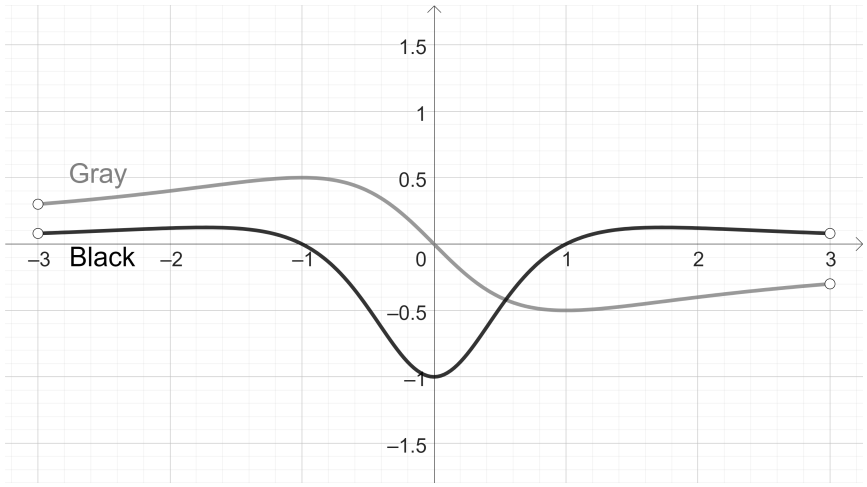
Solution: We search for x values for which the function is undefined, specifically $x = -1$ and $x = 1$. To prove that these are each vertical asymptotes, we consider one sided limits approaching the two values.

$$\lim_{x\to-1^+} \frac{-x}{x^2-1} = \frac{1}{0^-} \to -\infty$$

$$\lim_{x\to1^+} \frac{-x}{x^2-1} = \frac{-1}{0^+} \to -\infty$$

These limits show that there are vertical asymptotes at $x = -1$ and $x = 1$.

4. Below are two curves: "Gray" and "Black". Compute the following.



(a) (2 points) The "Gray" function evaluated at $x = 1$. *No need to show work.*

Solution: From the graph, our answer is $\frac{-1}{2}$.

(b) (2 points) The "Black" function evaluated at $x = 1$. *No need to show work.*

Solution: From the graph, our answer is 0.

(c) (3 points) The slope of the tangent to the "Gray" curve at $x = 1$. Draw this tangent line on the graph above.

Solution: The function is not changing at $x = 1$, so our answer is 0.

(d) (3 points) The slope of the tangent to the "Black" curve at $x = 1$. Draw this tangent line on the graph above. *An estimate is okay here but give a reasonable answer.*

Solution: The function is increasing slightly, so a reasonable answer is $\frac{1}{2}$.

(e) (6 points) One of these curves is the derivative graph of the other curve. In other words, one of these curves is $f(x)$ while the other curve is $f'(x)$. Which one is which? Explain in words how you know. *Hint: Your answers to the previous parts may help here.*

Solution: The function evaluation of the Black curve matches the tangent line slope of the Gray curve. This implies that Gray is $f(x)$ while Black is $f'(x)$.

5. Consider the following function, where m and b are unknown constants:

$$f(x) = \begin{cases} \sin(x), & x \leq 0 \\ mx + b, & 0 < x < \pi \\ \cos(x), & x \geq \pi \end{cases}$$

(a) (6 points) Explain in words why this function is definitely continuous for all $x \neq 0$ and $x \neq \pi$ no matter the choice of m and b.

Solution: Each piece of the function is continuous everywhere; $mx + b$ is a polynomial, while $\sin(x)$ and $\cos(x)$ are known to be continuous trig functions.

(b) (12 points) Find the values of m and b that make this function continuous everywhere.

Solution: We proceed using one sided limits approaching $x = 0$ and $x = \pi$ respectively.

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} \sin(x) = 0$$
$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} mx + b = b$$
$$f(0) = \sin(0) = 0$$

Because all three of these should be equal, we have $b = 0$.

$$\lim_{x \to \pi^-} f(x) = \lim_{x \to \pi^-} mx + 0 = \pi m$$
$$\lim_{x \to \pi^+} f(x) = \lim_{x \to \pi^+} \cos(x) = -1$$
$$f(\pi) = \cos(\pi) = -1$$

So, $\pi m = -1$ and $m = \frac{-1}{\pi}$.

6. Consider the function $f(x) = a\sqrt{x}$, where a is some unknown constant.

   (a) (12 points) **Using the limit definition for the derivative**, find $f'(1)$. Work done with derivative rules will not be graded.

   Solution: We use the definition in which $h \to 0$:

$$\begin{aligned}
\lim_{h\to 0} \frac{f(1+h) - f(1)}{h} &= \lim_{h\to 0} \frac{a\sqrt{1+h} - a\sqrt{1}}{h} \\
&= \lim_{h\to 0} a\frac{\sqrt{1+h} - 1}{h}\left(\frac{\sqrt{1+h}+1}{\sqrt{1+h}+1}\right) \\
&= a\lim_{h\to 0} \frac{(1+h) - 1}{h(\sqrt{1+h}+1)} \\
&= a\lim_{h\to 0} \frac{1}{\sqrt{1+h}+1} \\
&= \frac{a}{2}
\end{aligned}$$

   (b) (4 points) Find any value(s) of a that make the tangent line to the curve at $x = 1$ parallel to $y = x$.

   Solution: We set our previous answer to be the same as the slope of $y = x$, which is 1. From this, we solve:

$$\frac{a}{2} = 1 \implies a = 2$$

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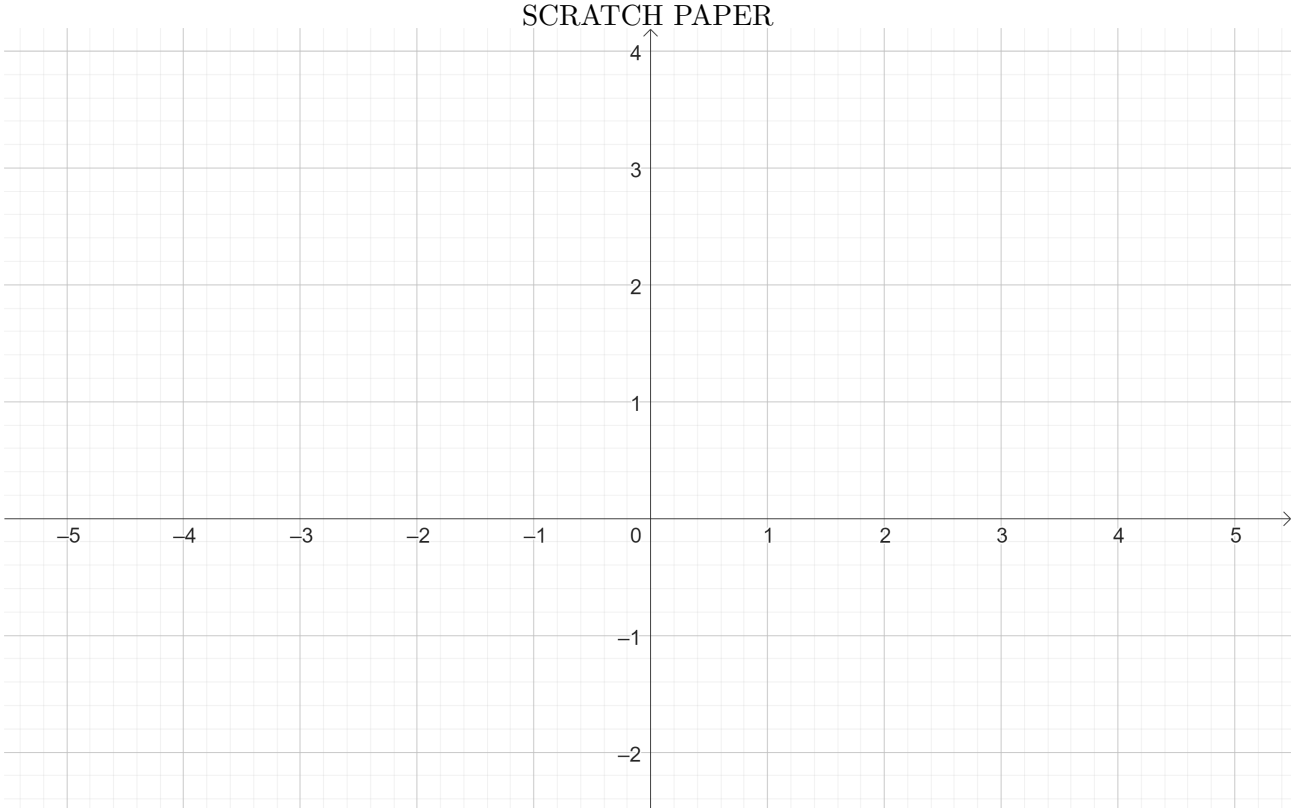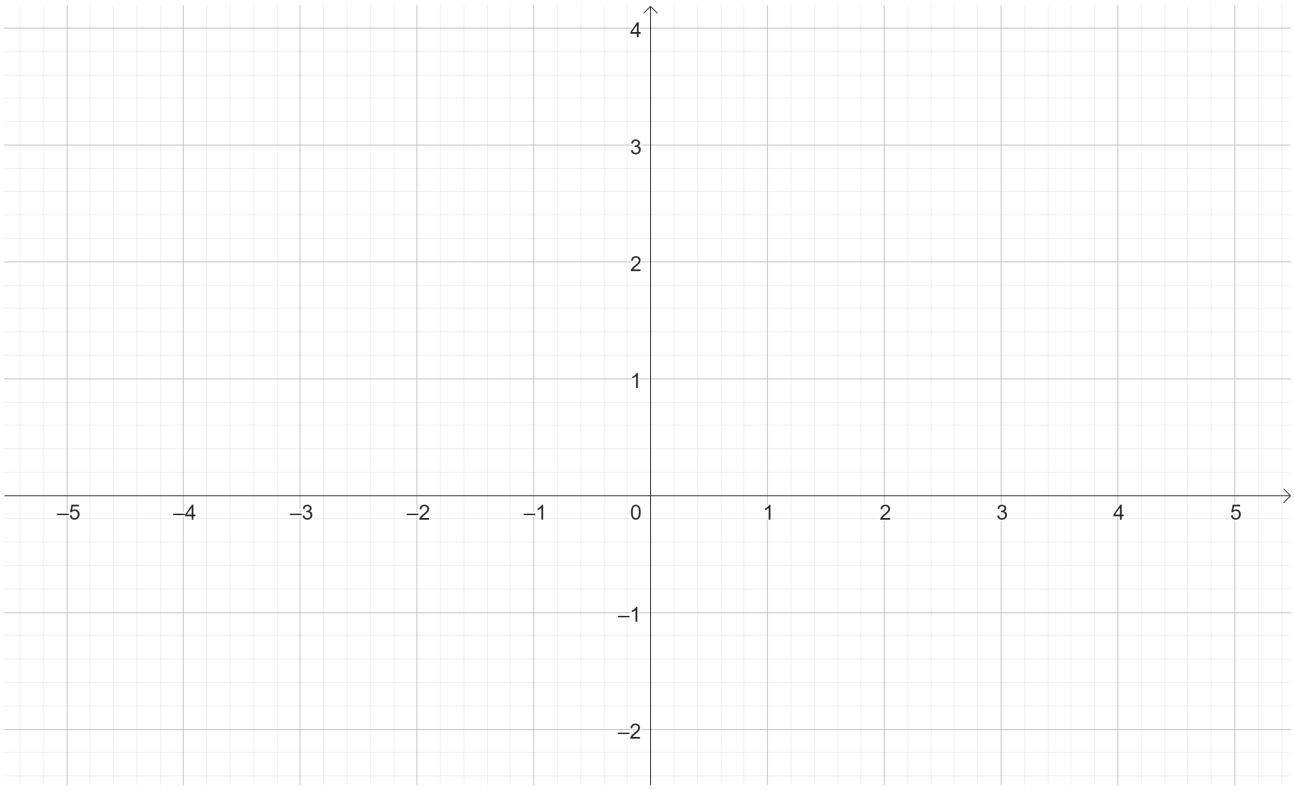
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